

Thesis:

Concrete built the modern world. Now it's destroying it. A growing chorus of architects argues we have to build differently with concrete or perhaps abandon it altogether.

Expected word count:

~3,500 words

Links to articles you've published elsewhere:**Pitch:**

Concrete enabled modern civilization. Behind water, it is the second most used substance on the planet. Today, most of humanity lives on islands built of concrete. It is the lifeblood of logistics – buildings, roads and bridges. It is also war – the Modern War Institute described it as “the most effective weapon on the modern battlefield.” Even the biggest religious and political icons on earth – like Christ the Redeemer and the Statue of Unity – are made from reinforced concrete. It's no surprise that this substance has fuelled so many visions of both utopias and dystopias.

Why this works: It colorfully introduces us to the concept the author will be exploring and why this material is so crucial to modern society. Colorful quotes and details make the piece engaging.

Concrete has become universal because it's affordable, relatively simple to make, and remarkably versatile. But it is also destroying the Earth. The carbon emissions of aviation (around 2.1% of all human-induced CO2 emissions) have rightly come under scrutiny in recent times, but they are vastly dwarfed by the cement industry (concrete's key ingredient), which is responsible for around 8%. In terms of humanity's control of nature, concrete has been like our nuclear bomb: enabling us to flatten lands, dam rivers, contain seas, and drive tunnels through mountains. And as concrete has covered the Earth, it has erased histories that lurked beneath.

As the population rises, the demand for concrete correlates. And yet, the great irony of this monolithic material is that while its destructive force is driving the climate crisis, it will also be a vital tool in protecting us. The more innovative concrete buildings are capable of withstanding earthquakes, hurricanes, floods and fires, and can even operate as carbon sinks. Around the world, coasts are being reinforced with concrete to protect against rising seas: 14% of the America's and 60% of China's coastline are now effectively concrete.

Why this works: The sentence, “And yet ... protecting us,” clearly sets out the key tension that will be featured in the piece and why we should care. Examples and data illustrate this point from a perspective that invites interest not solely from one region of the world.

I want to write a story that explores the past, present and future of concrete – our cultural fascination with it and fundamental reliance on it. I'm most intrigued by casting concrete as both villain and (hopefully) savior, if we follow the right path. After all, the difficulty of turning the world off concrete towers over that of meat eating, air travel, or plastic use. Vaclav Smil described the substance as one of the four indispensable pillars of modern life. And yet, we must look to change it.

Why this works: *The author explains their plan for the piece, why they are pursuing it in this way and, again, why anyone should care.*

There are new ideas everywhere. In Europe, the visionary architect Philippe Block is convinced that concrete is not the problem – it's how we use it. His geometric designs, inspired by the ancient origins of Roman concrete, are being tested on social housing projects around the world. 'Green concrete' exists, and I'll talk to its producers about their struggle to convince the world to use it. While in the Global South, South Africa's Noncrete has found a way of turning invasive plant species into biocement. And in Burkina Faso, the son of a tribe chief, Francis Kere, has become globally renowned for forgoing concrete altogether and using mud.

Why this works: *The author gives examples and details about possible solutions ("new ideas"), which is what Noema aims to focus on (rather than problems).*

There are lots of incredible historical characters, bold scientists and architects lurking within this topic that I would love to interview and shine a light on. There have been a few stories about concrete, like this one in [2019](#), but I would tackle it very differently, as an incredible amount has changed in the last two years.

Why this works: *Colorful characters help make a piece less dry — see the two interesting individuals the author mentions in the prior paragraph. The author has done their research, knows what else is out there and how this piece will push the subject beyond those existing stories into new territory.*